

Delivery Resource Planning System

Product Requirements Document (updated to match the current implementation)

1. Problem Statement

A delivery company receives a large number of orders every day but has a limited number of vehicles and delivery personnel available. Currently, deliveries are assigned manually, which leads to inefficient use of vehicles, unnecessary fuel consumption, delayed deliveries, and missed business opportunities.

This project builds a terminal-based system that helps the company plan its daily deliveries - deciding how the available vehicles and personnel should be used to handle incoming orders.

The company works like a porter service for e-commerce orders. The customer charge for an order is calculated by the system from the order's weight and distance (Rs20 per kg plus a per-km rate that depends on priority and vehicle class). Nobody types in a price. Every trip also costs money to run, at a per-km rate fixed by priority and vehicle type. Vehicles have fixed carrying capacities, and orders carry different priorities and deadlines.

The system must produce a daily delivery plan that makes the best use of limited resources - balancing delivery completion, cost, revenue, and profitability - with the overall objective of maximizing profit while respecting all resource and operational constraints.

2. Interface

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  DELIVERY RESOURCE PLANNING SYSTEM
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1. Add Order
2. View All Orders
3. Add Vehicle
4. Add Delivery Personnel
5. Generate Daily Delivery Plan
6. View Last Plan / Report
7. Exit
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Choose an option:
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Add Order prompts for:

- Order ID (entered; must be unique)
- Distance (km) - pickup to drop
- Weight (kg)
- Priority (1. Express 2. Standard)
- Deadline (hours from now)

Revenue is NOT entered. The system calculates it (see Section 3) and prints it when the order is added. An order is refused, and not stored, if any acceptance rule in FR12 fails.

Add Vehicle prompts for:

- Vehicle ID (entered; must be unique)

- Type (1. Bike 2. Van 3. Truck)

Max capacity and the per-km rates are built in for each type, so they are not asked for.

Add Delivery Personnel prompts for:

- Personnel ID (entered; must be unique)
- Name

(No vehicle assignment here - the system pairs personnel with vehicles itself when it generates the plan.)

Generate Daily Delivery Plan

No input needed. The system pairs each available vehicle with an available personnel, runs the allocation logic on all pending orders, and prints the plan - including which driver got which vehicle.

Report output (real output from the system for the sample orders below):

Vehicles: V1 Van (driver Rohit), V2 Bike (driver Anu). Orders (distance, weight, priority, deadline): O1 (5km, 3kg, Express, 2hr) = Rs335; O3 (8km, 12kg, Express, 3hr) = Rs680; O7 (6km, 8kg, Standard, 3hr) = Rs340; O9 (4km, 30kg, Standard, 2hr) = Rs820; O12 (40km, 5kg, Express, 1hr) = Rs2300.

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DELIVERY PLAN - SUMMARY
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Vehicle V1 (Van, driver: Rohit)
  -> Order 09 (Standard, 30kg, Rs820)
  -> Order 07 (Standard, 8kg, Rs340)
  Total load: 38/65 kg | Trip cost: Rs550.00

Vehicle V2 (Bike, driver: Anu)
  -> Order 03 (Express, 12kg, Rs680)
  -> Order 01 (Express, 3kg, Rs335)
  Total load: 15/15 kg | Trip cost: Rs715.00

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Orders Rejected (insufficient capacity/time/profit):
  012 (Express, 5kg) -- no vehicle had capacity/time
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Total Revenue      : Rs2175.00
Total Cost         : Rs1265.00
NET PROFIT        : Rs910.00
Orders Completed   : 4 / 5
=====
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3. Built-in Business Rules

These values are fixed in the code. Users never enter them.

3.1 Vehicle capacity and speed

Vehicle type	Max capacity	Average speed
Bike	15 kg	30 km/h
Van	65 kg	25 km/h
Truck	200 kg	20 km/h

3.2 Per-km rate (Rs per km)

The same table is used in two places: as the running cost of a trip (by the vehicle that carries the order and the order's priority) and as the per-km part of the customer charge (by the order's priority and its vehicle class, see 3.3).

Priority	Bike	Van	Truck
Standard	30	55	70
Express	55	80	95

3.3 Revenue (customer charge) per order

Revenue = (weight in kg x Rs20) + (distance in km x per-km rate).

The per-km rate is taken from the table in 3.2 using the order's priority and its vehicle class. The vehicle class is the smallest vehicle type that can carry the order's weight: up to 15 kg is Bike, over 15 and up to 65 kg is Van, over 65 and up to 200 kg is Truck. Orders over 200 kg cannot be carried and are refused.

Worked example: 5 km, 10 kg, Standard. Class is Bike, so revenue = $10 \times 20 + 5 \times 30 = \text{Rs}350$.

Because the running cost uses the same per-km rates, an order carried by a vehicle of its own class earns a profit equal to its per-kg charge (here $10 \text{ kg} \times \text{Rs}20 = \text{Rs}200$). If the order is carried by a larger vehicle, the running cost is higher and the profit shrinks or turns negative.

3.4 Order acceptance minimums

- Revenue must be at least Rs100.
- Pickup-to-drop distance must be at least 200 m (0.2 km).
- Weight must not exceed 200 kg (the largest vehicle).
- Distance exactly 0.2 km and revenue exactly Rs100 are accepted (the checks are 'less than').

4. Functional Requirements

ID	Requirement
FR1	The system shall allow the user to add an order with: ID, distance, weight, priority (Express/Standard), and deadline. Revenue is calculated by the system (FR11), not entered.
FR2	The system shall allow the user to register a vehicle with an ID and a type (Bike/Van/Truck). Max carrying capacity and per-km rates are fixed by type (Section 3).
FR3	The system shall allow the user to register delivery personnel with an ID and name, independent of any vehicle.
FR4	The system shall generate a daily delivery plan that: automatically pairs each vehicle used in the plan with an available delivery personnel (no manual driver-vehicle assignment); assigns orders so no vehicle exceeds its carrying capacity; assigns orders only to a vehicle that can deliver them before their deadline; and aims to maximize total profit = (sum of revenue from completed orders) - (sum of trip cost for vehicles used).
FR4a	If there are more vehicles than available personnel (or vice versa), the system shall use only as many vehicle-personnel pairs as it can form, and shall report any vehicles left idle due to no personnel being available (or personnel left idle due to no vehicle being available).

ID	Requirement
FR5	When total order demand exceeds available capacity, the system shall rank orders by: (1) priority, Express before Standard; (2) revenue, higher first, so an order that pays more is accepted before one that pays less; (3) profit per kg; (4) deadline urgency; (5) entry order. Priority always wins outright over revenue.
FR6	The system shall reject orders that cannot be fit into any vehicle without violating capacity or deadline constraints, and shall report each rejected order along with the reason.
FR7	The system shall calculate and display total revenue, total operating cost, and net profit for the generated plan.
FR8	The system shall display, per vehicle: assigned driver, list of orders carried, total load used vs. capacity, and trip cost.
FR9	The system shall validate all numeric inputs (distance, weight, deadline) and reject zero, negative, or non-numeric values with a re-prompt. Priority and vehicle type must be a valid menu choice, and IDs must be unique.
FR10	The system shall allow viewing all currently entered orders/vehicles/personnel before a plan is generated.
FR11	The system shall calculate each order's revenue as (weight x Rs20) + (distance x per-km rate for the order's priority and vehicle class), as defined in Section 3.3.
FR12	The system shall refuse, and not store, an order if: its distance is under 200 m; its weight is over 200 kg; or its calculated revenue is under Rs100. A message states the reason.
FR13	When more than one vehicle can take an order, the system shall assign it to the vehicle with the lowest cost to carry it. If two or more vehicles cost exactly the same, the order goes to the first vehicle-driver pair, which is the first delivery person in the personnel list (pairing is in ID order).
FR14	After allocation, the system shall check each vehicle's route. While trip cost exceeds the route's revenue, it shall remove the order that loses the most money on that vehicle and reject it with the reason 'unprofitable at route level'.

5. Test Cases

ID	Scenario	Input	Expected Output
TC1	Add a valid order	ID=O1, 5km, 3kg, Express, deadline 2hr	Order stored; revenue calculated as Rs335 (3x20 + 5x55); appears in order list
TC2	Add order with negative weight	weight = -5	System rejects input, re-prompts
TC3	Add a valid vehicle	ID=V1, type=Van	Vehicle stored with built-in capacity 65 kg and built-in rates
TC4	Plan with enough capacity for all orders	3 small Standard orders (5km/3kg, 4km/2kg, 3km/4kg), 1 Van	All 3 orders assigned, profit = total revenue - total cost
TC5	Total weight exceeds vehicle capacity	8 orders x 10kg (80kg), 1 Van (65kg)	6 orders assigned, 2 rejected; assigned weight does not exceed 65 kg
TC6	Deadline cannot be met by any vehicle	100km Express order, deadline 0.5hr, one Bike	Order rejected, reason 'no vehicle had capacity/time'
TC7	Plan with zero orders	No orders added	System reports 'No orders to plan' instead of crashing
TC8	Plan with zero vehicles	Orders exist, no vehicles	All orders rejected with reason 'no vehicles available'

ID	Scenario	Input	Expected Output
TC9	Express vs Standard compete for the same space	1 Express and 1 Standard order, space for only one	Express is ranked first and gets the space, even if the Standard order pays more (FR5)
TC10	Multiple vehicles, orders distributed	5 orders (30,30,30,20,10 kg = 120kg), 2 Vans (65kg each)	Orders split across both vans; at least 3 assigned; both vans used
TC11	More vehicles than personnel	3 vehicles, 2 personnel	2 pairs formed in ID order; 3rd vehicle reported idle (no personnel available)
TC12	Plan with zero personnel	Vehicles and orders exist, no personnel	All orders rejected with reason 'no personnel available'
TC13	Order weight exactly equals remaining capacity	65kg order, Van with 65kg free	Order accepted (boundary)
TC14	Deadline exactly equals delivery time	25km on a Van (25 km/h), deadline 1hr	Order accepted as just in time
TC15	Deliverable order with revenue below its trip cost	50km Standard order, revenue Rs50, Van (cost 50x55 = Rs2750)	Order rejected, reason 'unprofitable at route level'
TC16	Duplicate ID entered	Existing Order ID re-entered	System says 'ID already exists' and re-prompts; existing record untouched
TC17	Non-numeric input for a numeric field	weight = 'five'	System re-prompts without crashing
TC18	Deadline zero or negative	deadline = -1 or 0	System re-prompts; the order is never stored with an invalid deadline
TC19	Vehicle-driver pair idle due to incompatible orders	Bike (15kg) + Van, 2 drivers; orders of 20kg and 30kg	Bike pair reported idle with reason 'no compatible orders'
TC20	Capacity and rates are not user-entered	Add a Bike, a Van and a Truck	Capacities are 15, 65, 200 kg; Standard rates 30/55/70; Express rates 55/80/95
TC21	Loss-making order sharing a vehicle	O1: 5km Rs300 (cost Rs275). O2: 20km Rs300 (cost Rs1100). One Van	O2 rejected 'unprofitable at route level'; O1 stays; route cost does not exceed route revenue
TC22	Two orders tie exactly	Orders A and B identical in priority, revenue, profit per kg, deadline	Order entered first (A) is ranked first; result repeatable
TC23	Large-scale stress test	500 orders, 5 Vans	Plan completes, valid report produced, no crash
TC24	Revenue below Rs100	1km, 3kg, Standard (revenue Rs90)	Order not accepted, not stored
TC25	Revenue exactly Rs100	2km, 2kg, Standard (2x20 + 2x30)	Order accepted (boundary)
TC26	Distance under 200 m	distance = 0.15km	Order not accepted, not stored
TC27	Weight over 200 kg	weight = 250kg	Order not accepted, not stored
TC28	Revenue calculation	10km/10kg Standard; 10km/20kg Express; 10km/100kg Standard; 10km/15kg Express	Rs500; Rs1200; Rs2700; Rs850
TC29	Higher paying order is accepted first	Two Standard 60kg orders paying Rs500 and Rs300, one Van (65kg)	Rs500 order assigned, Rs300 order rejected
TC30	Equal cost goes to the first delivery person	Two identical Vans V1, V2 paired with P1, P2; one order	Order goes to V1 (paired with P1)

ID	Scenario	Input	Expected Output
TC31	Cheapest vehicle preferred	Bike and Van both able to carry one order	Order goes to the Bike (Rs30/km < Rs55/km for Standard)
TC32	Profit on a matching-class vehicle	5km, 10kg Standard order (Rs350) on a Bike (cost Rs150)	Profit is Rs200 (10kg x Rs20)

6. Edge Cases

ID	Description
EC1	Plan generated with zero orders entered: system prints 'No orders to plan' and exits gracefully.
EC2	Plan generated with zero vehicles: all orders rejected with reason 'no vehicles available'.
EC3	Plan generated with zero personnel: no vehicle can be paired, so all orders are rejected with reason 'no personnel available'.
EC4	Equal numbers of vehicles and personnel, but one vehicle cannot carry any pending order (for example a 15 kg Bike when every order is heavier): that pair sits idle with reason 'no compatible orders', which is different from 'no personnel available'.
EC5	Order weight exactly equals a vehicle's remaining capacity (for example 65 kg into a Van with exactly 65 kg free): must be accepted (guards against a '<' vs '<=' bug).
EC6	Order deadline exactly equals the time needed to deliver it: must be accepted as just in time.
EC7	Capacity and per-km rates are built in and always above 0, so a zero capacity or zero cost vehicle cannot be created.
EC8	An order fits capacity and deadline but its revenue is less than the cost it adds to the trip: rejected on profit grounds.
EC9	Route-level check: the system re-checks each vehicle's whole route after allocation. Because cost and revenue are both sums over orders, a route can only be unprofitable if at least one order on it loses money on that vehicle; the loss-making orders are removed worst first.
EC10	Two orders tie on priority, revenue, profit per kg and deadline: the one entered first wins, so results are reproducible.
EC11	An Express order has lower revenue or profit per kg than a competing Standard order: Express still wins the space, because priority is the first ranking key.
EC12	Duplicate Order ID / Vehicle ID / Personnel ID: rejected with a re-prompt, never overwritten.
EC13	Non-numeric input where a number is expected: re-prompts without crashing.
EC14	Deadline of 0 or negative hours: re-prompted at input time, never passed to planning.
EC15	500+ orders against a handful of vehicles: terminates in reasonable time. A greedy ranking (FR5) is used instead of a brute-force optimal solver because optimal bin packing is NP-hard.
EC16	Revenue of exactly Rs100 is accepted; anything below it is refused.
EC17	Distance of exactly 0.2 km passes the distance rule. It is still refused if its revenue is under Rs100; a heavier order (for example 5 kg Express, Rs111) is accepted.
EC18	Weight over 200 kg: refused at entry because no vehicle can carry it.
EC19	Two or more vehicles cost exactly the same for an order: the first vehicle-driver pair (lowest personnel ID) gets it.
EC20	A small order (for example Bike class) placed on a larger vehicle because the cheaper vehicle is full: it costs more to carry, so its profit shrinks and it may be dropped by the route check.